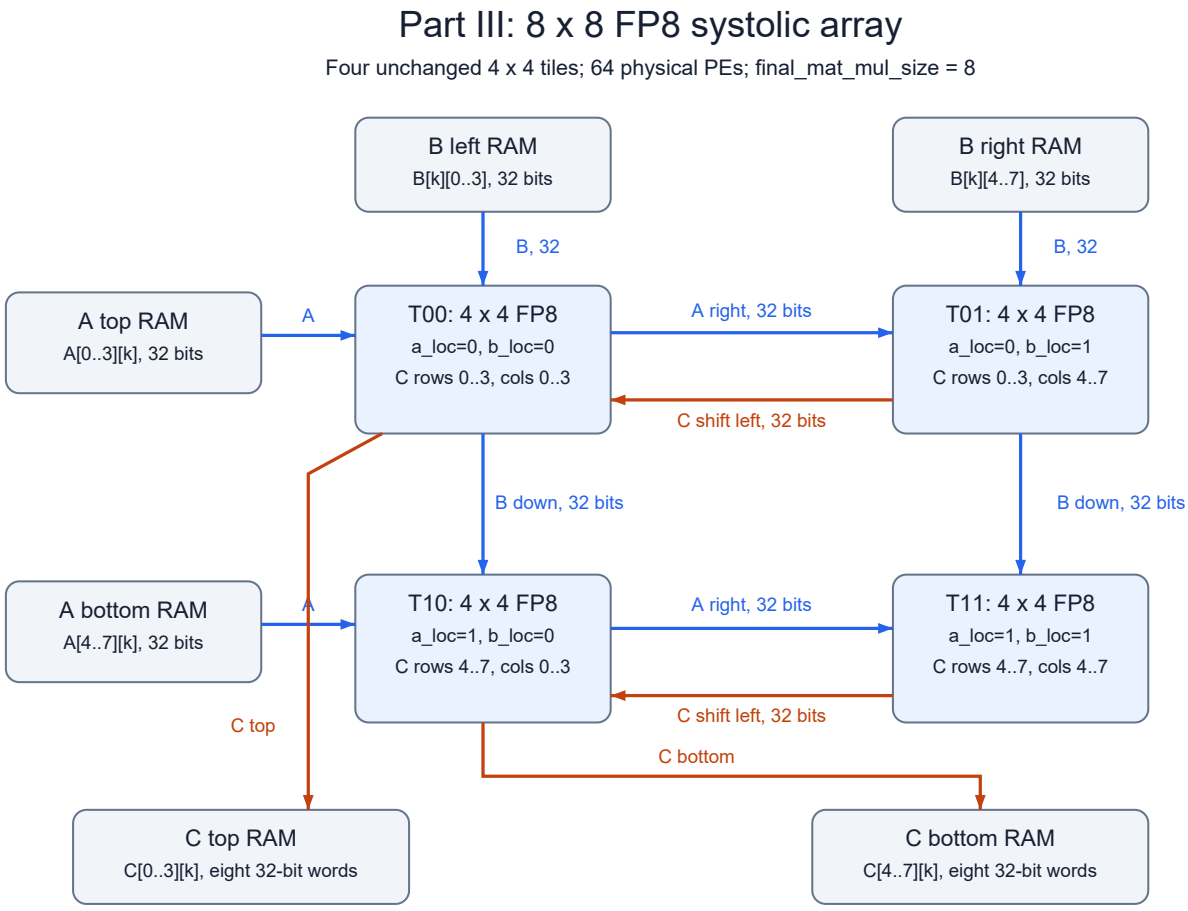
Lab 3 - Part III

8 x 8 FP8 systolic matrix multiplier

Implemented and verified on the Apporto course server, October 21, 2026. Scope: Part III only. Parts I and II supply unchanged FP8 arithmetic and 4 x 4 tiles. No Part IV APB work is included.

Microarchitecture



Every tile receives final_mat_mul_size=8 and masks=1111. a_loc is tile row and b_loc is tile column (0 or 1). A moves right and B moves down through 32-bit links. Each PE computes a full eight-term dot product; the opposite-direction C links serialize completed columns, rather than forwarding arithmetic partial sums.

32-bit bank	Word k, byte r (r=0..3)
A top / A bottom	A[r][k] / A[r+4][k]
B left / B right	B[k][r] / B[k][r+4]
C top / C bottom	C[r][k] / C[r+4][k]

Byte 0 is bits 7:0. The six RAMs retain the supplied synchronous, read-first behavior. Two 32-bit banks per matrix hold eight words each. Full-word host loading is blocked while busy. The four five-bit sticky exception vectors are combined with bitwise OR.

Control, completion and required analysis

The inherited tile done threshold is fixed at 17, but the N=8 output latch correctly occurs at core count 26. The new controller holds run across the early done pulse. That pulse also changes the inherited C address counter, so the new core generates C bank addresses from the actual column-write count, base and stride. Part II matmul.v and MAT_MUL_SIZE=4 remain unchanged.

The RAM wrapper accepts an idle rising start, samples configuration, clears all tiles during one preparation clock, and then runs. Exactly eight column transfers write both C banks. Wrapper done is retained until reset or the next accepted start. Flags survive completion and clear during preparation for the next operation.

All-ones event	Time (ns)	Interpretation
Accepted start edge	290	TB logs after NBA at 291 ns
Preparation / first run edge	310 / 330	First core run uses pre-edge count 0
Top/left / lower-right reads	370 / 450	Qualified synchronous responses
Legacy done observed	690	Ignored, before final capture
Capture completed C	850	Pre-edge core count 26
Eight C write edges	870, 890, 910, 930, 950, 970, 990, 1010	Eight words committed per bank
Wrapper done edge / TB observes	1030 / 1040	Status follows final committed write

Required answer	Measured/derived result
Peak read bandwidth	128 bits/cycle = 16 bytes/cycle (2 A + 2 B banks, each 32 bits)
Peak write bandwidth	64 bits/cycle = 8 bytes/cycle (2 C banks)
Cycles to compute and shift out	35 inclusive core run edges: 27 through capture, then 8 output transfers. Including wrapper preparation, 36 periods from accepted host start to final write; done follows after 37 periods.
Useful MAC evaluations	8 x 8 x 8 = 512 scalar MAC steps; 64 physical PE/MAC units.

All four qualified read streams overlap at core counts 6..9. Internal PE forwarding and external loading are excluded from these RAM bandwidth numbers. At 50 MHz simulation frequency, peak read/write rates are 800/400 MB/s; at the 100 MHz synthesis target they are 1.6/0.8 GB/s. Peak and whole-operation average rates differ.

Verification and matrix cases

Actual VCS W-2024.09-SP2-3 sequence: all ones, deterministic varied signed data with nonzero bases/nonunit strides, finite overflow, then clean all-ones restart after overflow. All four operations pass: **256 output bytes checked**. Each operation checks every C byte through the synchronous RAM host port, all input bank loads and qualified read words, 32 lane-0 tile-forwarding samples, exactly eight C writes per bank, flags, stable completion and a finite timeout.

Case 1: A and B are 8 x 8 all-ones matrices (FP8 30). The actual C RAM contains 8 in all 64 positions (FP8 60); flags=00000. Case 3 uses A[7][7]=8 (60) and B[7][7]=2 (40), with other operands zero. C[7][7]=+Inf (70), other outputs zero; flags=10001. The final clean run clears the exception without external reset.

Case 2 below uses seed 320268. Cells show hexadecimal FP8 encodings; decimal values are in the supplied MATRICES.md. Expected results use the unchanged Part I exact-rational oracle, rounding each multiply and add separately in streamed k=0..7 order. Flags=00000. A base=17, stride=3; B base=43, stride=2; C base=101, stride=5.

Matrix A

row	0	1	2	3	4	5	6	7
0	00	b0	b0	a0	30	00	a0	20
1	30	30	a0	a0	a0	30	30	b0
2	20	30	20	20	a0	20	20	30
3	b0	30	a0	b0	a0	20	a0	00
4	20	00	30	20	20	a0	b0	a0
5	30	00	30	30	20	a0	00	a0
6	20	30	30	a0	a0	00	b0	00
7	30	b0	30	20	a0	00	20	30

Matrix B

row	0	1	2	3	4	5	6	7
0	00	30	30	a0	20	20	00	b0
1	20	b0	30	b0	30	b0	b0	20
2	30	a0	30	00	20	20	a0	a0
3	a0	00	b0	20	a0	30	b0	30
4	b0	00	20	b0	00	30	30	00
5	a0	30	b0	00	00	a0	00	a0
6	20	30	a0	20	30	b0	b0	30
7	20	30	b0	20	b0	20	00	00

Matrix C (actual RAM matched every expected byte)

row	0	1	2	3	4	5	6	7
0	c2	38	b4	90	c2	3c	4c	b0
1	10	34	34	b4	4c	ce	bc	90
2	3c	34	a0	10	28	a8	c6	20
3	20	bc	00	a8	10	c8	10	00
4	90	c0	46	b4	00	46	20	b4
5	00	a0	42	a8	30	44	b0	90
6	3c	c0	4a	b8	30	90	a0	c0
7	38	48	b0	40	a8	40	b0	b0

Core-only synthesis comparison

Metric	FP8 4 x 4	FP8 8 x 8
Cell area (library units)	33,263.044845	124,066.023235
Leaf / sequential cells	11,045 / 955	40,929 / 3,522
Full-interface setup slack (ns)	4.936644	2.114082
Setup period estimate (ns)	5.063356	7.885918
Estimated full-interface MHz	197.50	126.81
Register-to-register estimate, MHz	254.42	240.25
Hold violations	461	1,704
Maximum capacitance violations	3,657	14,669

Both runs: Design Compiler V-2023.12-SP5-1, gscl45nm typical, compile_ultra, target period 10 ns, uncertainty/clock transition 0.10 ns, input/output delay 1 ns, input transition 0.10 ns, and output load 0.01 pF. The 8 x 8 run synthesizes matmul_8x8 only; no wrapper, memories or testbench. The 4 x 4 numbers come from the provided Part II raw reports; they were not freshly rerun for this comparison.

The estimate uses T_est = 10 ns - setup slack and frequency_MHz = 1000 / T_est under fixed I/O budgets. It is a pre-layout setup estimate, not routed Fmax or a clock sweep. Area is about 3.73 times the provided 4 x 4 area; library area units were not established as physical square micrometres.

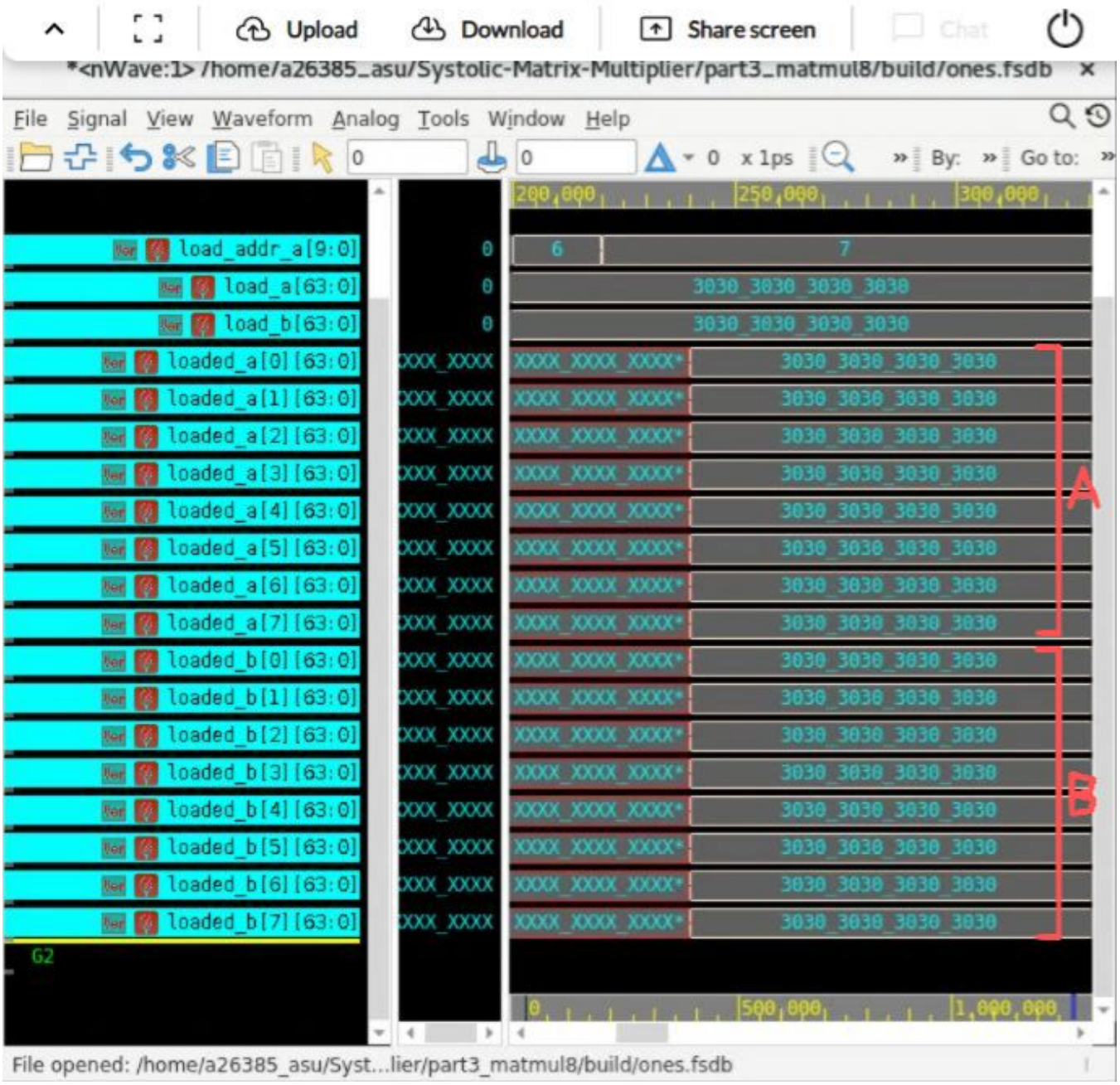
Interfaces differ: Part II exposes tile locations/final size as unconstrained inputs. The required 8 x 8 composition specializes those constants and adds output sequencing. The 8 x 8 worst path is pe_reset input to a PE accumulator, reflecting reset fanout; the 4 x 4 worst path is final_mat_mul_size input to a C address register.

Timing limitation: both have residual hold and capacitance violations. Worst 8 x 8 hold slack is -0.066914 ns. Full timing closure is not claimed. The mapped 8 x 8 has zero black-box area; check_design reports only the expected constant divide-by-zero flag output. check_timing reports no unconstrained endpoints, loops or missing input delays, and warns about a high-fanout net.

Partner contributions

Partner	Contribution

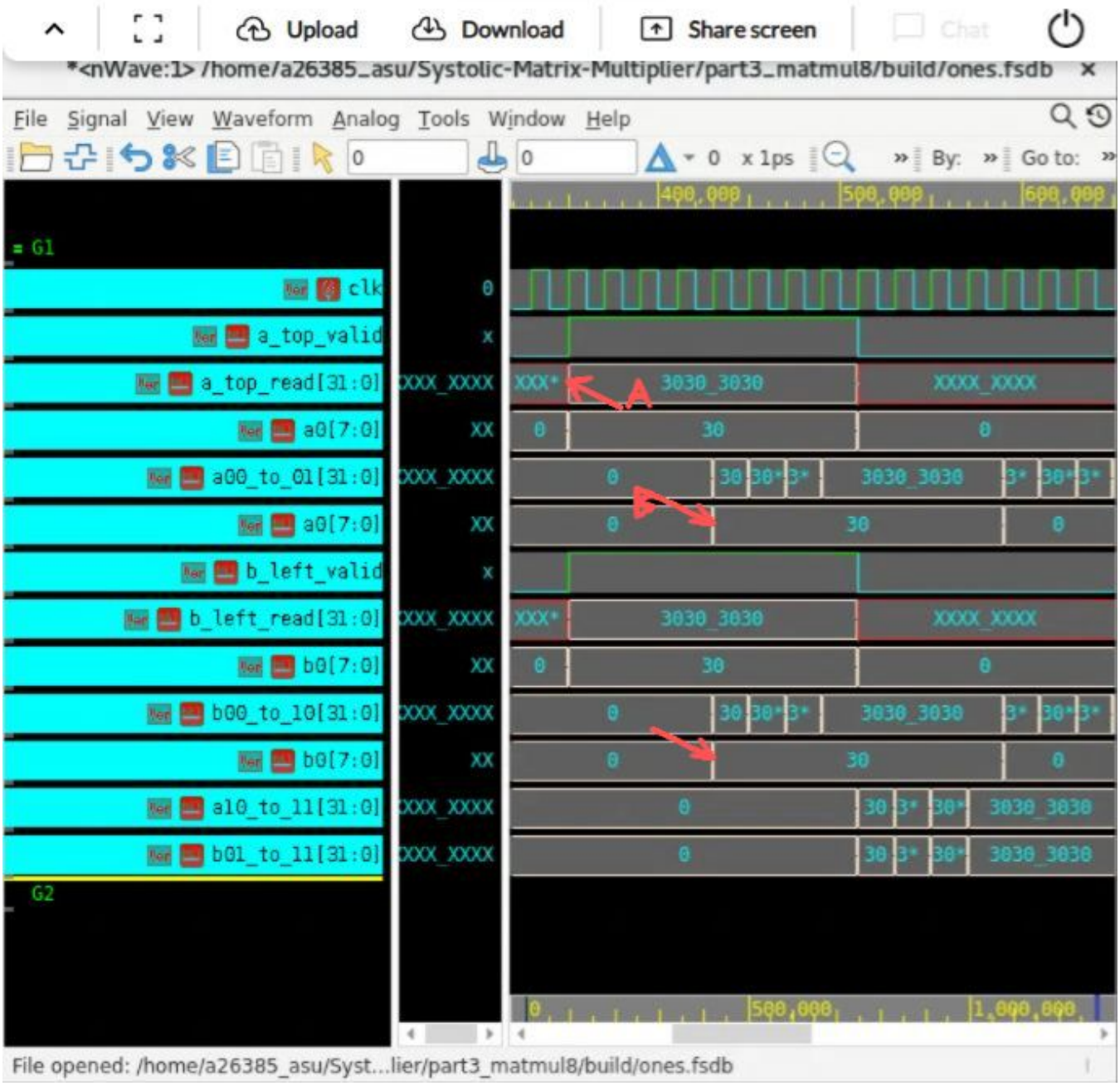
Input matrices in RAM



Digital-pen brackets A and B mark eight packed columns from the two A banks and eight packed rows from the two B banks. Each 64-bit word concatenates two actual 32-bit banks. All bytes are 30 (FP8 1). loaded_a and loaded_b snapshot actual RAM after loading, rather than copying expected arrays. Time ruler is in ps (200,000 ps = 200 ns).

Source: actual Part III build/ones.fsdb in the course Verdi nWave viewer. The unmarked PNG is preserved alongside this pen copy. Tcl scripts document signals and time intervals. The narrow value column follows the zero-time cursor; read actual values from the plotted traces in the displayed time window.

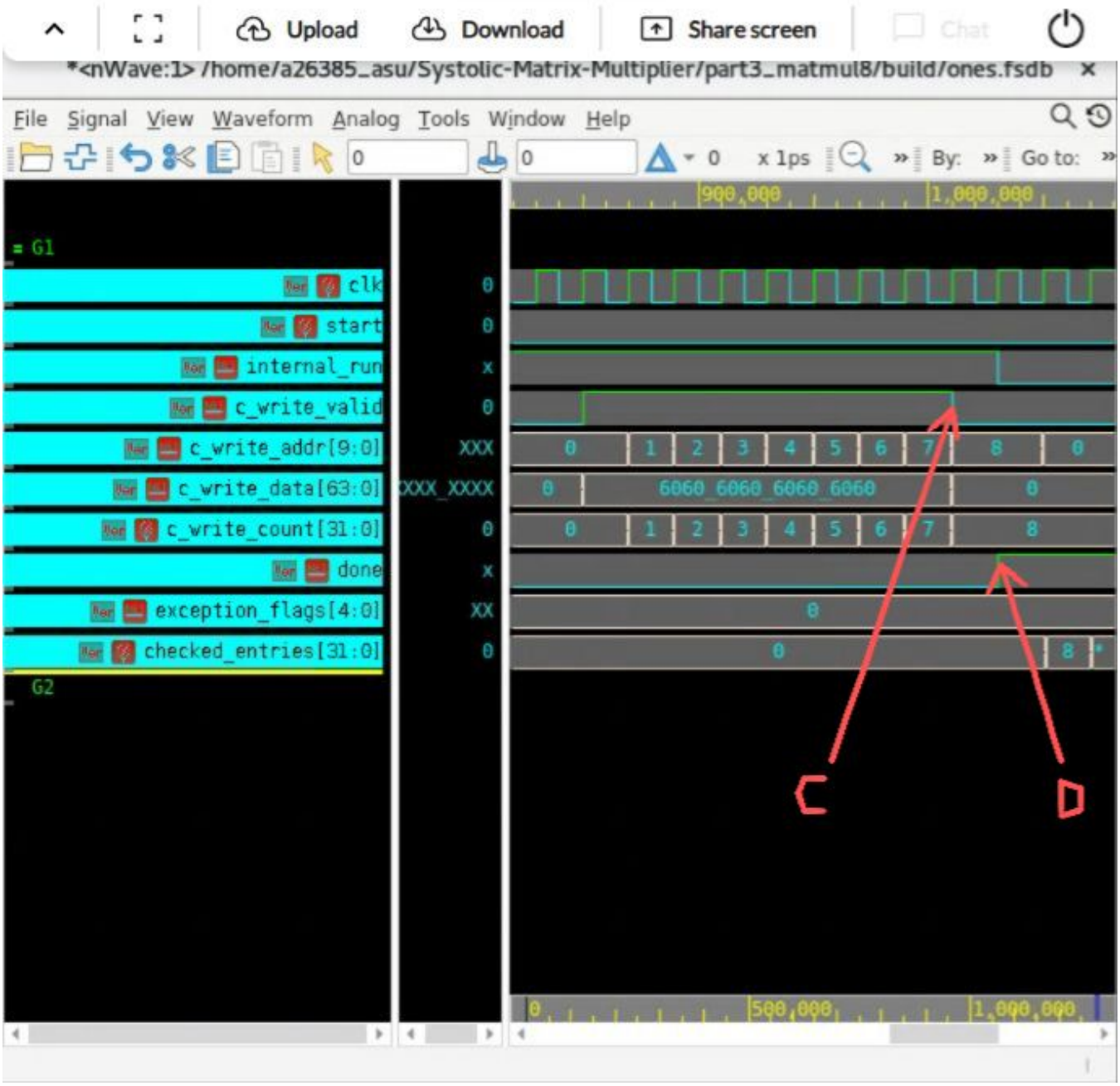
RAM reads and transfer to neighboring tiles



Digital-pen arrow A marks the qualified RAM response at the array edge. B arrows mark the operands reaching the next tile after four registered PE steps. A travels left to right and B top to bottom. A connected output/input boundary is the same wire; the four-cycle delay occurs inside the source tile. Time ruler is in ps.

Source: actual Part III build/ones.fsdb in the course Verdi nWave viewer. The unmarked PNG is preserved alongside this pen copy. Tcl scripts document signals and time intervals. The narrow value column follows the zero-time cursor; read actual values from the plotted traces in the displayed time window.

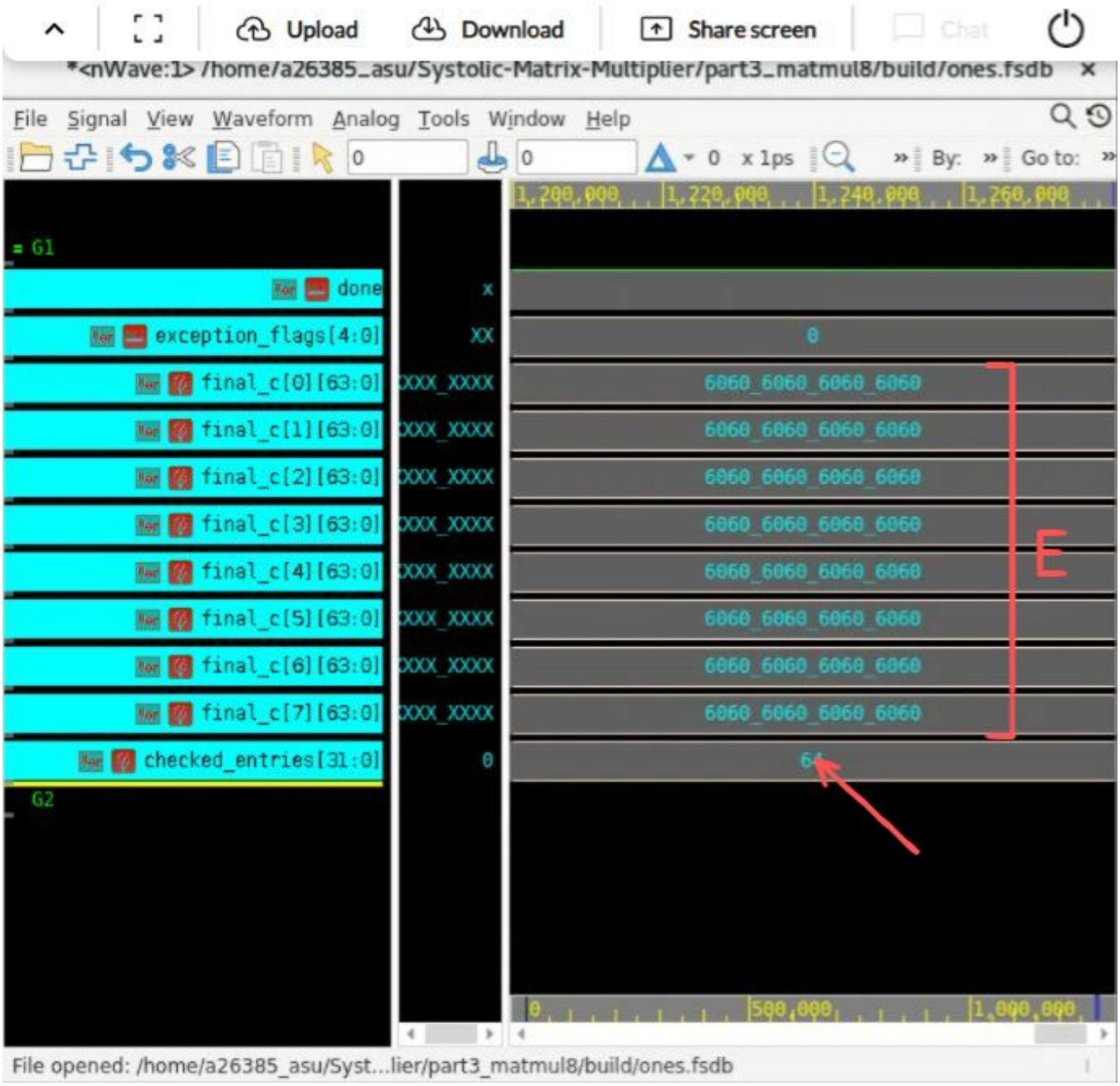
Eight output writes and completion



Digital-pen arrow C marks the final committed C write at 1010 ns, when the write enable drops after the edge and the completed column count becomes eight. Arrow D marks wrapper done at 1030 ns. Both output banks write together. Data is 6060606060606060, encoding eight FP8 results equal to 8.0. Time ruler is in ps.

Source: actual Part III build/ones.fsdb in the course Verdi nWave viewer. The unmarked PNG is preserved alongside this pen copy. Tcl scripts document signals and time intervals. The narrow value column follows the zero-time cursor; read actual values from the plotted traces in the displayed time window.

All 64 actual output elements



Digital-pen bracket E identifies the eight 64-bit actual C RAM readback words. Every byte is 60 (FP8 8), and checked_entries=64 confirms all 64 comparisons. These final_c values were captured through synchronous output RAM host reads, not taken from the expectation arrays. Time ruler is in ps.

Source: actual Part III build/ones.fsdb in the course Verdi nWave viewer. The unmarked PNG is preserved alongside this pen copy. Tcl scripts document signals and time intervals. The narrow value column follows the zero-time cursor; read actual values from the plotted traces in the displayed time window.